



ASSIGNMENT 2

VLSI FINAL PROJECT

SURE TRUST

Physical Design Analysis of Counter using OpenLane / Sky130

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Assignment Title	Assignment 2 - VLSI Final Project - SURE Trust

Professional formatted submission copy. All original assignment content is retained; front matter, charts, flow diagram, index, and important notes are added for presentation quality.



Executive Summary

This report documents the complete physical design analysis performed for the counter design using OpenLane and the Sky130 HD standard-cell library. **The original assignment material, screenshots, commands, tables, observations, GUI tasks, and conclusions are preserved without deletion.** The following pages provide a professional front section with a quick project snapshot, index, flow diagram, visual charts, and important notes before the full original report content.

Parameter	Details
Design topic	Counter implementation and physical design analysis
PD flow	Synthesis, Floorplan, Placement, CTS, Routing, STA/Signoff
Technology	Sky130 HD Standard Cell Library
Timing result	Setup and hold slack remain positive at the normal analysis stages
Maximum frequency result	Estimated Fmax is approximately 680 MHz; setup timing becomes the limiting constraint
Power trend	Total power increases from floorplan to routing as physical and parasitic effects become more realistic

Important retention note: No original paragraphs, tables, screenshots, command snippets, or observations have been intentionally removed. The original report starts after the professional visual summary section.

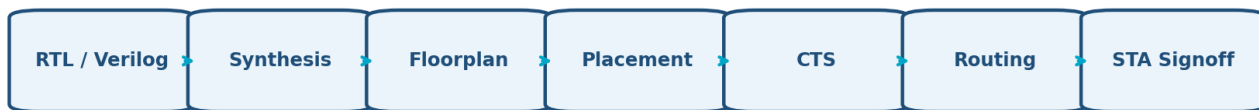


Index

S.No.	Section	Description
1	Objective	Timing performance analysis of the counter design across PD stages
2	OpenLane flow and STA extraction	Commands for synthesis, floorplan, placement, CTS, routing and slack extraction
3	Timing slack analysis	Worst setup and hold slack trend with observations
4	Area and utilization analysis	Standard-cell area, core/die dimensions, utilization and CTS buffer impact
5	Power analysis	Internal, switching, leakage and total power report generation
6	Clock uncertainty and timing derates	Clock uncertainty, early derate, late derate and timing margin explanation
7	Routing length analysis	Clock route length, signal route length and total routed length contribution
8	IR drop analysis	Worst IR drop and percentage calculation
9	Maximum frequency analysis	Frequency sweep, timing violation point and Fmax conclusion
10	GUI and layout observations	Core/die size, physical-only cells, buffers, flops and site rows
11	32-bit counter section	Timing, power, area, routing, IR drop and maximum frequency observations
12	Complete original report	Original assignment content retained in full after the visual summary



Professional Visual Summary



OpenLane ASIC Physical Design Flow used for the counter implementation

Figure 1: OpenLane physical design flow used in the report

Key Charts Generated from Report Data

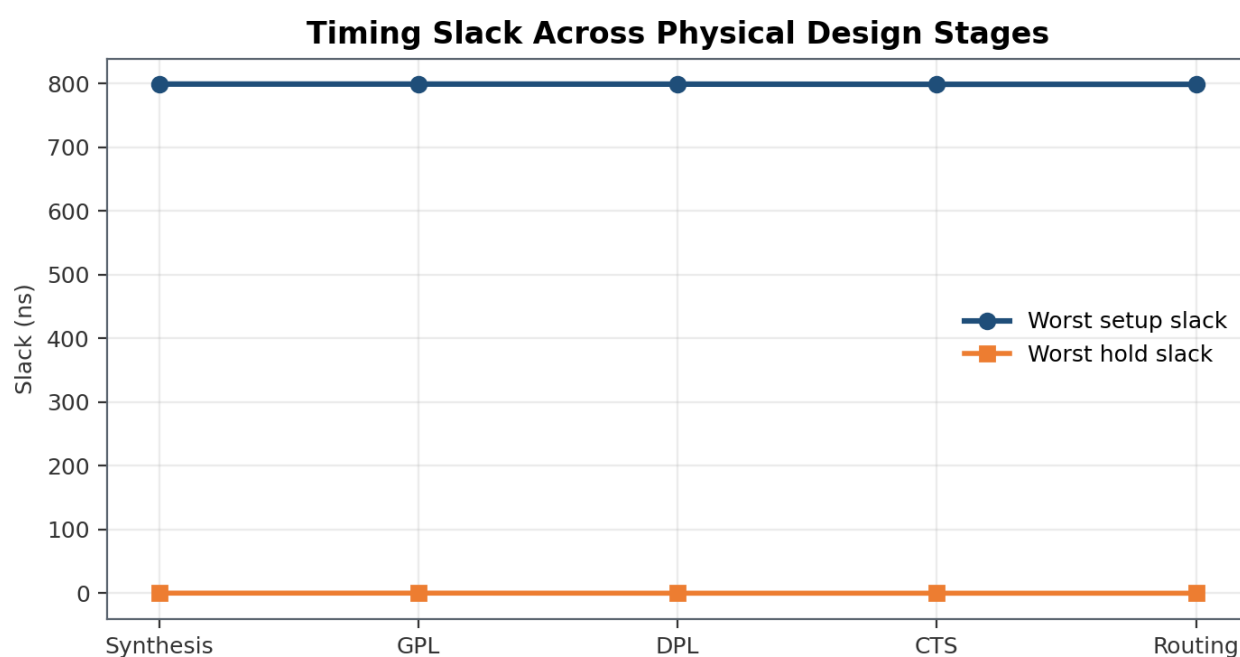


Figure 2: Worst setup and hold slack across physical design stages



Total Power Trend Across PD Stages

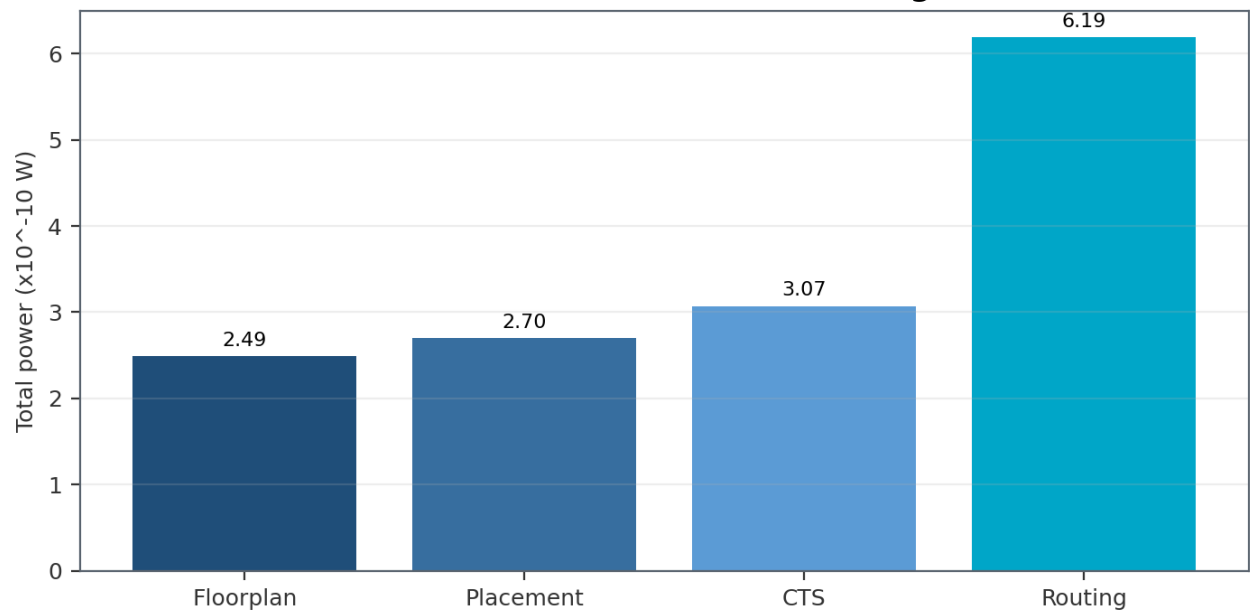


Figure 3: Total power trend from floorplan to routing

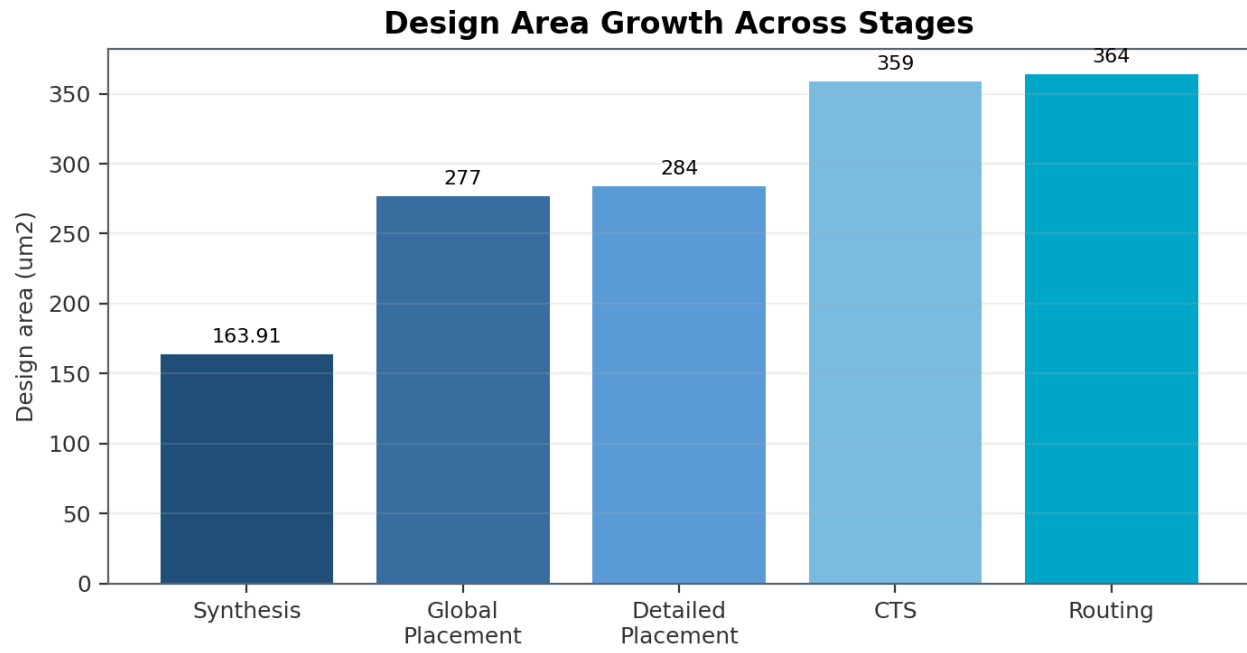


Figure 4: Design area growth through implementation stages

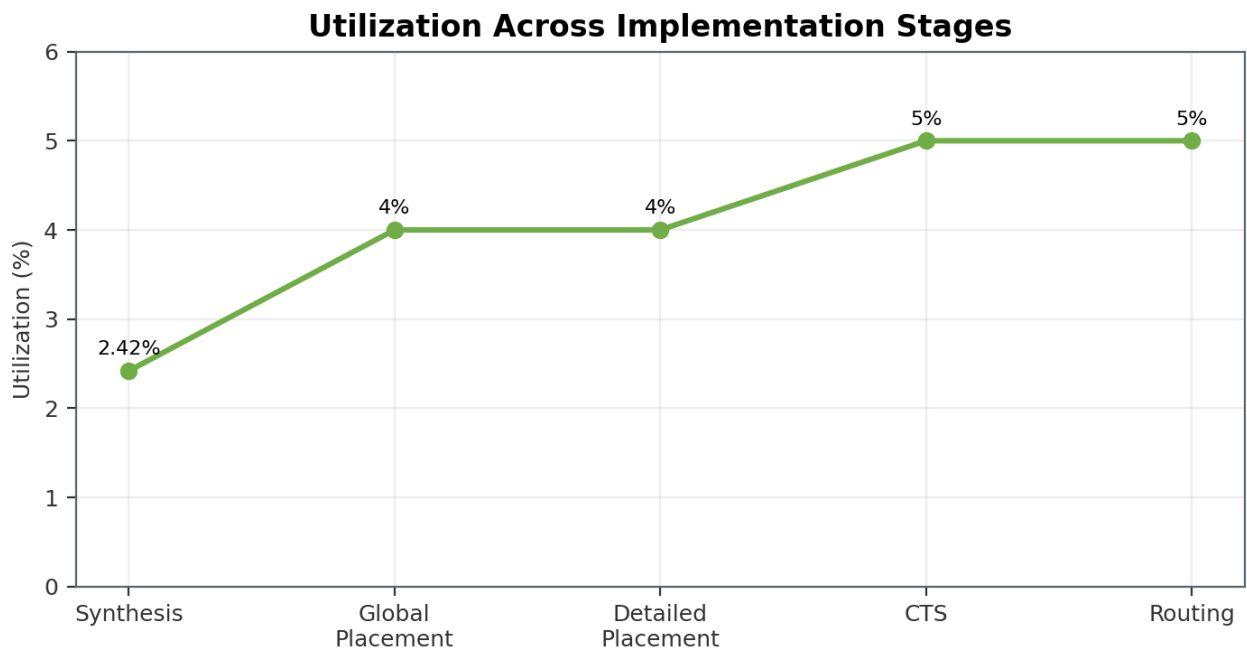


Figure 5: Utilization trend across stages

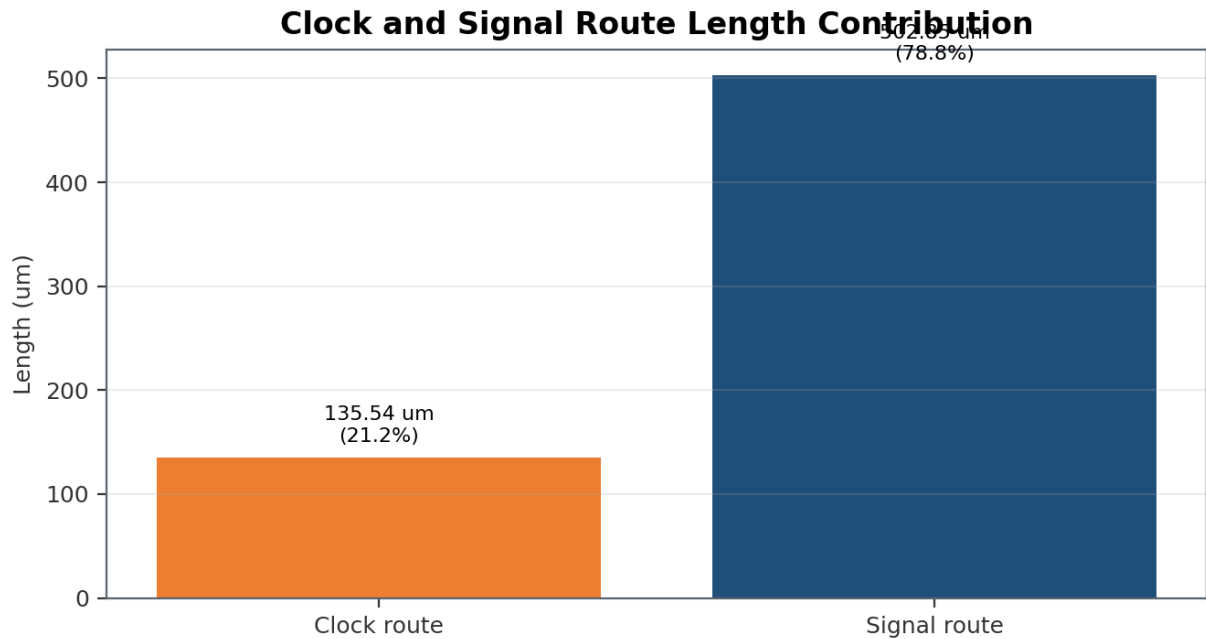


Figure 6: Clock and signal route length contribution

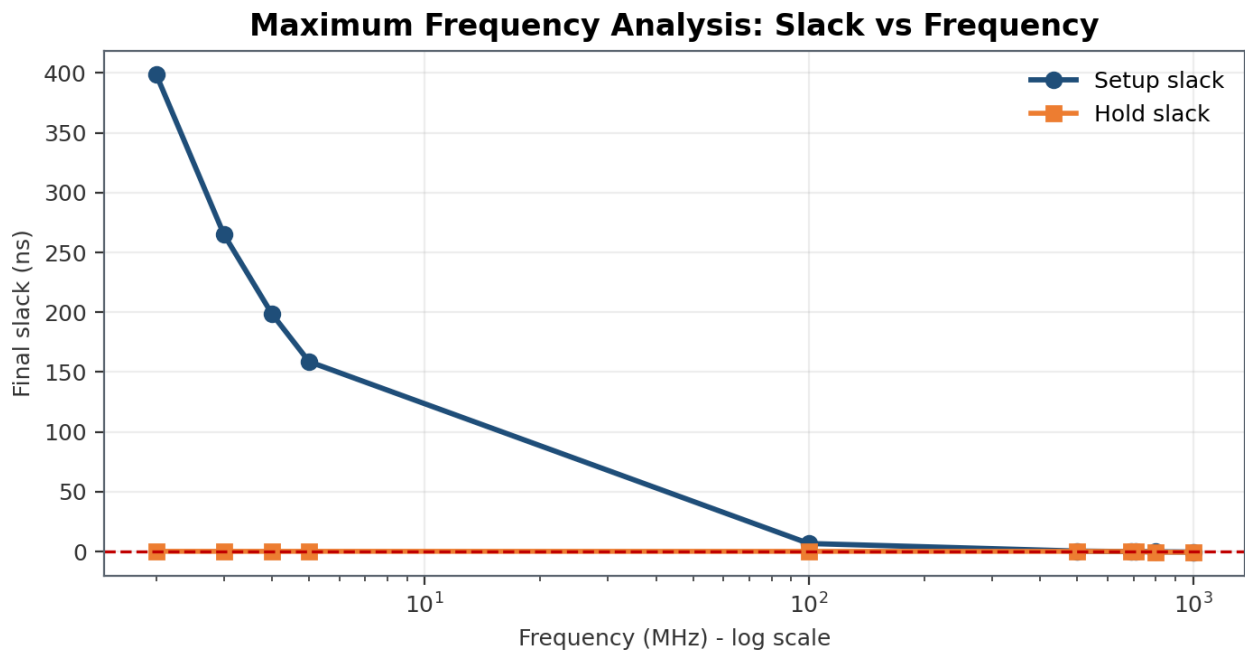


Figure 7: Frequency versus setup/hold slack for maximum frequency analysis

Important Notes for Professional Submission

- All positive setup slack values indicate that setup timing is met for the normal physical design flow stages.
- All positive hold slack values indicate that hold timing is met for the normal physical design flow stages.
- Setup slack and hold slack reduce slightly as the flow progresses because placement, CTS and routing introduce more realistic physical information.
- CTS is important because clock buffers and clock-tree insertion can affect skew, insertion delay, area and power.
- Routing includes actual interconnect and parasitic effects, making post-routing timing and power results more realistic.



1. Objective

To analyze the timing performance of the 4-bit counter at different stages of the Physical Design flow and observe how setup and hold slack vary as more realistic physical information is introduced.

2. Steps Followed to Extract Worst Slack

Step 1: Run the complete PD flow

```
prep -design 4-bitCounter  
run_synthesis  
run_floorplan  
run_placement  
run_cts  
run_routing
```

Step 2: Extract slack from STA logs

For each stage, the STA log was searched using:

Synthesis

```
grep "worst slack" logs/synthesis/2-sta.log
```

Global Placement (GPL)

```
grep "worst slack" logs/placement/8-gpl_sta.log
```

Detailed Placement (DPL)

```
grep "worst slack" logs/placement/11-dpl_sta.log
```

CTS

```
grep "worst slack" logs/cts/13-cts_sta.log
```

Routing

```
grep "worst slack" logs/routing/20-grt_sta.log
```

Stage	Worst Setup Slack (ns)	Worst Hold Slack (ns)
Synthesis	799.17	0.32
Global Placement (GPL)	799.16	0.34
Detailed Placement (DPL)	799.05	0.21
CTS	798.80	0.20
Routing	798.77	0.17

Observation 1

All setup slack values are positive.

Setup Slack > 0

Therefore:

No setup timing violations



Observation 2

All hold slack values are positive.

Hold Slack > 0

Therefore:

No hold timing violations

Observation 3

Setup slack gradually decreases.

799.17

↓

799.16

↓

799.05

↓

798.80

↓

798.77

Reason:

- Placement introduces actual cell locations.
- CTS inserts clock buffers.

• Routing introduces wire resistance and capacitance.

These effects increase path delay slightly.

Observation 4

Hold slack also decreases.

0.34

↓

0.21

↓

0.20

↓

0.17

Reason:

As physical effects become more accurate, hold timing margins reduce slightly.

Stage	Area (μm^2)	Evidence
Synthesis	163.9072	1-synthesis.AREA_0.stat.rpt
Floorplan	Core Area = 6761.08 μm^2 (88.78 × 76.16)	3-initial_fp_core_area.rpt
Placement	Not explicitly reported	No area report generated
CTS	Not explicitly reported	CTS inserted 3 clock buffers
Routing	Not explicitly reported	No area report generated



Area Observations

Observation 1

The synthesized design contains:

10 standard cells

Area = $163.9072 \mu\text{m}^2$

This is the baseline standard-cell area of the design.

Observation 2

The floorplan allocated:

Core Width = $88.78 \mu\text{m}$

Core Height = $76.16 \mu\text{m}$

Core Area $\approx 6761.08 \mu\text{m}^2$

Die Width = $100 \mu\text{m}$

Die Height = $100 \mu\text{m}$

Die Area = $10000 \mu\text{m}^2$

This shows the design occupies only a small fraction of the available chip area.

Observation 3

CTS log confirms:

Total number of Buffers Inserted: 3

Therefore, additional standard cells were inserted into the design during CTS.

Observation 4

No CTS-area or routing-area report was generated by the current OpenLane configuration, so the exact post-CTS and post-routing standard-cell areas could not be extracted directly.

Steps to Generate Power Report at Each Stage

Step 1: Enter OpenLane Directory

```
cd ~/Desktop/Openlane/OpenLane
```

Step 2: Start Docker Container

```
make mount
```

Step 3: Launch OpenROAD

```
openroad
```



Step 4: Read Technology LEF

```
read_lef /openlane/designs/4-bitCounter/runs/RUN_3June/tmp/merged.nom.lef
```

If the merged LEF is stored elsewhere, use that path.

Step 5: Read Liberty File

Find liberty file:

```
find /openlane/pdks -name "*.lib" | grep sky130_fd_sc_hd | head
```

Example:

```
read_liberty /openlane/pdks/sky130A/libs.ref/sky130_fd_sc_hd/lib/sky130_fd_sc_hd__tt_025C_1v80.lib
```

Step 6: Load Stage Database

Floorplan Power

```
read_db results/floorplan/up_counter.odb
```

Placement Power

```
read_db results/placement/up_counter.odb
```

CTS Power

```
read_db results/cts/up_counter.odb
```

Routing Power

```
read_db results/routing/up_counter.odb
```

Power Table

Stage	Internal Power (W)	Switching Power (W)	Leakage Power (W)	Total Power (W)
Floorplan	8.13×10^{-14}	5.02×10^{-14}	2.48×10^{-10}	2.49×10^{-10}
Placement	1.07×10^{-13}	4.07×10^{-14}	2.70×10^{-10}	2.70×10^{-10}
CTS	1.32×10^{-13}	4.45×10^{-14}	3.07×10^{-10}	3.07×10^{-10}
Routing	1.33×10^{-13}	4.46×10^{-14}	6.19×10^{-10}	6.19×10^{-10}



The increase in leakage power across stages was:

- Floorplan: $2.48 \times 10^{-10} \text{ W}$
- Placement: $2.70 \times 10^{-10} \text{ W}$
- CTS: $3.07 \times 10^{-10} \text{ W}$
- Routing: $6.19 \times 10^{-10} \text{ W}$

For a real digital circuit:

Total Power = Dynamic Power + Leakage Power

where

Dynamic Power = Switching + Internal

Percentage Increase in Total Power

Floorplan → Placement

$$\frac{2.70 - 2.49}{2.49} \times 100$$

≈ 8.43%

Placement → CTS

$$\frac{3.07 - 2.70}{2.70} \times 100$$

≈ 13.70%

CTS → Routing

$$\frac{6.19 - 3.07}{3.07} \times 100$$

≈ 101.63%

Floorplan → Routing

$$\frac{6.19 - 2.49}{2.49} \times 100$$

≈ 148.6%

Analysis for Report



Floorplan Stage

- Power estimation is based on an early physical view.
- Interconnects are not fully characterized.
- Total power = $2.49 \times 10^{-10} \text{ W}$.

Placement Stage

- Standard cells are assigned physical locations.
- More accurate wire estimation is available.
- Total power increases to $2.70 \times 10^{-10} \text{ W}$.

CTS Stage

- Clock buffers and clock-tree elements are inserted.
- Additional cells increase leakage power.
- Total power rises to $3.07 \times 10^{-10} \text{ W}$.

Routing Stage

- **Actual routing and parasitic effects are included.**
- Leakage estimation becomes more realistic.
- Total power reaches $6.19 \times 10^{-10} \text{ W}$.

• Steps to Find Clock Uncertainty and Timing Derates

- **Step 1: Open Terminal**

- Go to the OpenLane directory:
- `cd /home/bhanu-sharma/Desktop/Openlane/OpenLane`

- **Step 2: Find Clock Uncertainty**

- Search all files in the run directory for clock uncertainty:
- `grep -Ri "clock_uncertainty" \`
`/home/bhanu-sharma/Desktop/Openlane/OpenLane/designs/4-bitCounter/runs/RUN_3June`
- Output obtained:
- `set_clock_uncertainty 0.2500 clk`
- Therefore,
- Clock Uncertainty = 0.25 ns

- **Step 3: Find Timing Derates**

- Search all files in the run directory for timing derates:
- `grep -Ri "timing_derate" \`
`/home/bhanu-sharma/Desktop/Openlane/OpenLane/designs/4-bitCounter/runs/RUN_3June`



Clock Uncertainty and Derate Table

Stage	Clock Uncertainty (ns)	Early Derate	Late Derate
Floorplan	0.25	0.95	1.05
Placement	0.25	0.95	1.05
CTS	0.25	0.95	1.05
Routing*	0.25	0.95	1.05

* Routing typically inherits the same constraints since no separate routing SDC with different values was found.

Observation

1. The clock uncertainty remained constant at **0.25 ns** throughout all physical design stages.
2. The timing derate values also remained unchanged:
 - Early Derate = **0.95**
 - Late Derate = **1.05**
3. The early derate of **0.95** reduces delays by 5%, representing the best-case timing scenario.
4. The late derate of **1.05** increases delays by 5%, representing the worst-case timing scenario.
5. These derates provide a timing margin to account for process, voltage, and temperature (PVT) variations and on-chip variation (OCV).
6. Since the same uncertainty and derate values were applied throughout the flow, timing comparisons between Floorplan, Placement, CTS, and Routing remain consistent.

Utilization Table

Stage	Design Area (μm^2)	Utilization (%)
Synthesis	163.91	2.42*
Global Placement	277	4
Detailed Placement	284	4
CTS	359	5
Routing	364	5

Observations

1. Area Increased After Placement

163.91 $\rightarrow$ 277 $\rightarrow$ 284 μm^2

Reason:

- Buffer insertion
- Gate resizing
- Timing optimization by resizer



2. Largest Increase Occurred During CTS

284 → 359 μm^2

Reason:

CTS inserted 3 clock buffers

These additional clock cells increase the design area.

Net	Length (μm)
net1	94.78
net3	89.84
net5	80.32
net2	73.68
net4	70.38
Clk	56.99
clknet_0_clk	38.99
clknet_1_0__leaf_clk	27.18
clknet_1_1__leaf_clk	12.38
...	...

Step 1: Calculate Clock Route Length

Clock-related nets are:

clk
clknet_0_clk
clknet_1_0__leaf_clk
clknet_1_1__leaf_clk

Total clock length:

56.99
+ 38.99
+ 27.18
+ 12.38

= 135.54 μm



Step 2: Calculate Total Routed Length

Sum of all nets:

94.78
+89.84
+80.32
+73.68
+70.38
+56.99
+38.99
+27.18
+14.82
+14.22
+12.38
+11.72
+10.54
+10.46
+10.32
+5.84
+5.60
+5.19
+2.62
+2.52

= 638.39 μm

Step 3: Signal Route Length

Signal Length

=

Total Length – Clock Length

=

638.39 – 135.54

=

502.85 μm

Assignment Table

Parameter	Length (μm)
Clock Route Length	135.54
Signal Route Length	502.85
Total Routed Length	638.39



Observation 2

The clock network contributes:

135.54 μm

which is:

$$(135.54 / 638.39) \times 100 \\ \approx 21.2\%$$

of the total routing.

Observation 3

Signal routing contributes:

502.85 μm

which is:

$$\approx 78.8\%$$

of the total routed length.

Step 1: Worst IR Drop

$$\text{Worstcase IR drop} = 2.86 \times 10^{-10} \text{ V}$$

Step 2: IR Drop Percentage

Formula:

$$\text{IR Drop \%} = (\text{Worst IR Drop} / \text{Supply Voltage}) \times 100$$

Supply Voltage:

$$\text{VDD} = 1.8 \text{ V}$$

Calculation:

$$(2.86 \times 10^{-10} / 1.8) \times 100$$

$$= 1.59 \times 10^{-8} \%$$



Therefore:

Parameter	Value
Worst IR Drop	2.86×10^{-10} V
Supply Voltage	1.8 V
Worst IR Drop %	1.59×10^{-8} %

Observation

The IR drop is extremely small.

Reason:

- Very small design (4-bit counter)
- Only a few standard cells
- Very low current consumption
- Short power distribution paths

Task : Increasing the Frequency to maximum value till timing violation occurs :

Final Conclusion – Maximum Frequency Analysis of 4-Bit Counter

The 4-bit synchronous counter was implemented through the complete OpenLane ASIC flow, including synthesis, floorplanning, placement, clock tree synthesis (CTS), routing, and timing signoff. To determine the maximum operating frequency, the clock period was progressively reduced and the design was re-implemented for each frequency target.

Timing analysis showed that the design comfortably met setup and hold requirements at lower frequencies (1 MHz to 500 MHz). As the clock frequency increased, the available setup slack gradually decreased due to tighter timing constraints. Hold violations were observed during synthesis, placement, and CTS stages; however, these were successfully repaired during post-routing timing optimization through automatic buffer insertion by the OpenLane resizer.

A detailed investigation was performed at higher frequencies. At 690 MHz, the final routed design reported a setup slack of -0.03 ns, indicating a setup timing violation, while hold timing remained positive. Similar and larger setup violations were observed at 710 MHz, 800 MHz, and 1000 MHz. Since timing closure was achieved at 500 MHz but failed at 690 MHz, the maximum achievable operating frequency of the implemented design is estimated to be approximately **680 MHz** (clock period ≈ 1.47 ns).

An important observation is that timing degradation became significant after Clock Tree Synthesis due to clock insertion delay and skew. Although post-routing optimization successfully fixed hold violations, setup timing ultimately became the limiting factor at higher frequencies. The results demonstrate the impact of CTS, routing parasitics, and timing optimization on overall design performance.



Final Result:

- Maximum Achievable Frequency (Fmax): $\approx$ **680 MHz**
- Corresponding Minimum Clock Period: $\approx$ **1.47 ns**
- Limiting Constraint: **Setup Timing**
- **Hold Violations: Successfully fixed during post-routing timing optimization**
- Design Status at 680 MHz: **Timing Closure Achieved**
- Design Status at 690 MHz and above: **Timing Violations Observed**

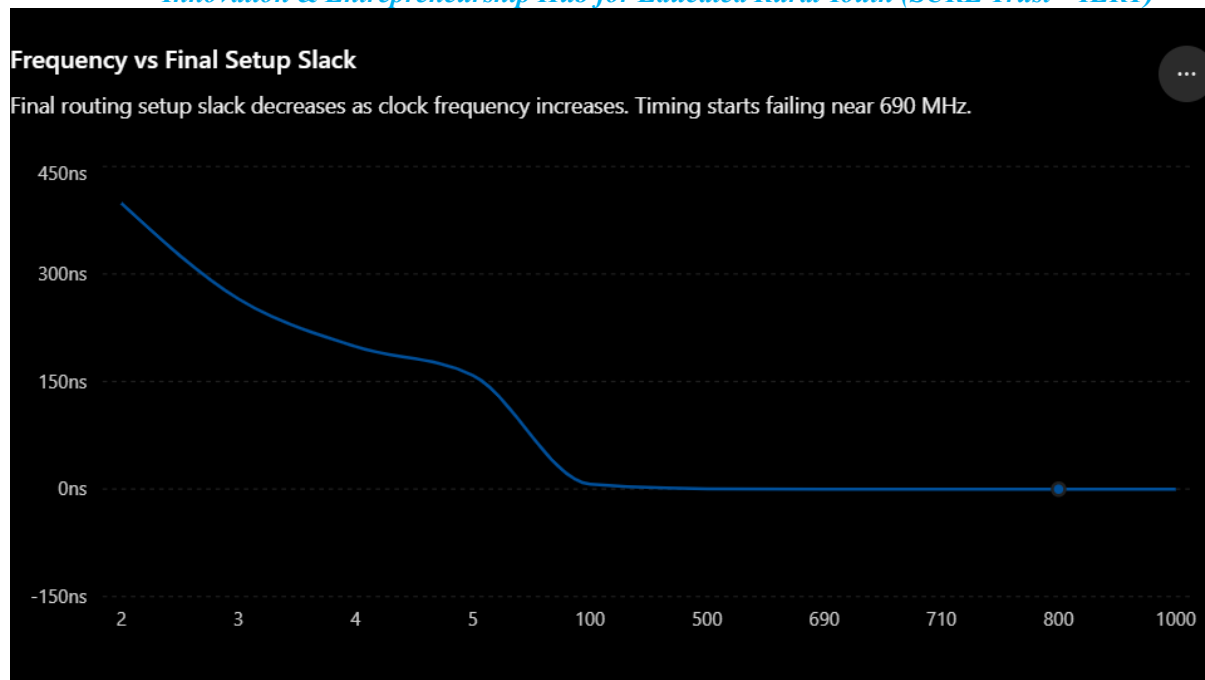
Maximum Frequency Analysis Summary

Frequency (MHz)	Clock Period (ns)	Final Setup Slack (ns)	Final Hold Slack (ns)	Status
2	500.00	+398.74	+0.20	PASS
3	333.33	+265.14	+0.20	PASS
4	250.00	+198.74	+0.20	PASS
5	200.00	+158.74	+0.20	PASS
100	10.00	+6.74	+0.17	PASS
500	2.00	+0.41	+0.17	PASS
690	1.449	-0.03	+0.08	FAIL (Setup)
710	1.408	-0.06	+0.07	FAIL (Setup)
800	1.250	-0.19	-0.47	FAIL (Setup & Hold)
1000	1.000	-0.39	-0.52	FAIL (Setup & Hold)

Graph: Frequency vs Setup Slack

Frequency vs Final Setup Slack

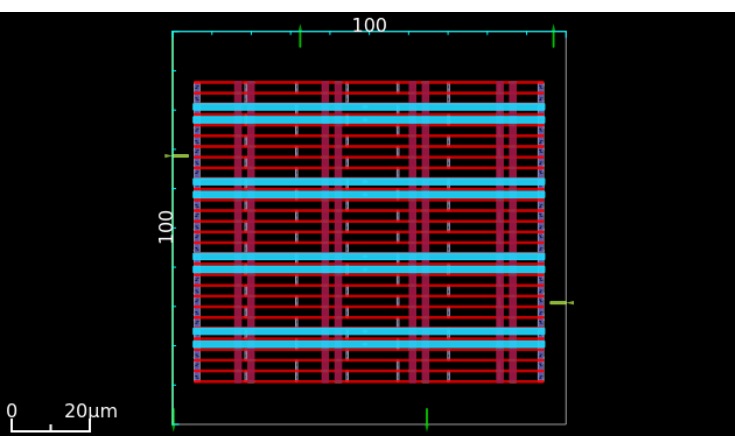
Final routing setup slack decreases as clock frequency increases. Timing starts failing near 690 MHz.



- ❑ Setup slack decreases continuously as clock frequency increases.
- ❑ The design comfortably meets timing up to **500 MHz**.
- ❑ At **690 MHz**, setup slack becomes **-0.03 ns**, indicating the first setup timing violation.
- ❑ Hold violations appearing after CTS are repaired during post-routing timing optimization up to about 710 MHz.
- ❑ Beyond **800 MHz**, both setup and hold timing constraints fail.
- ❑ CTS is the most critical stage where timing degradation is first observed due to clock insertion delay and skew.

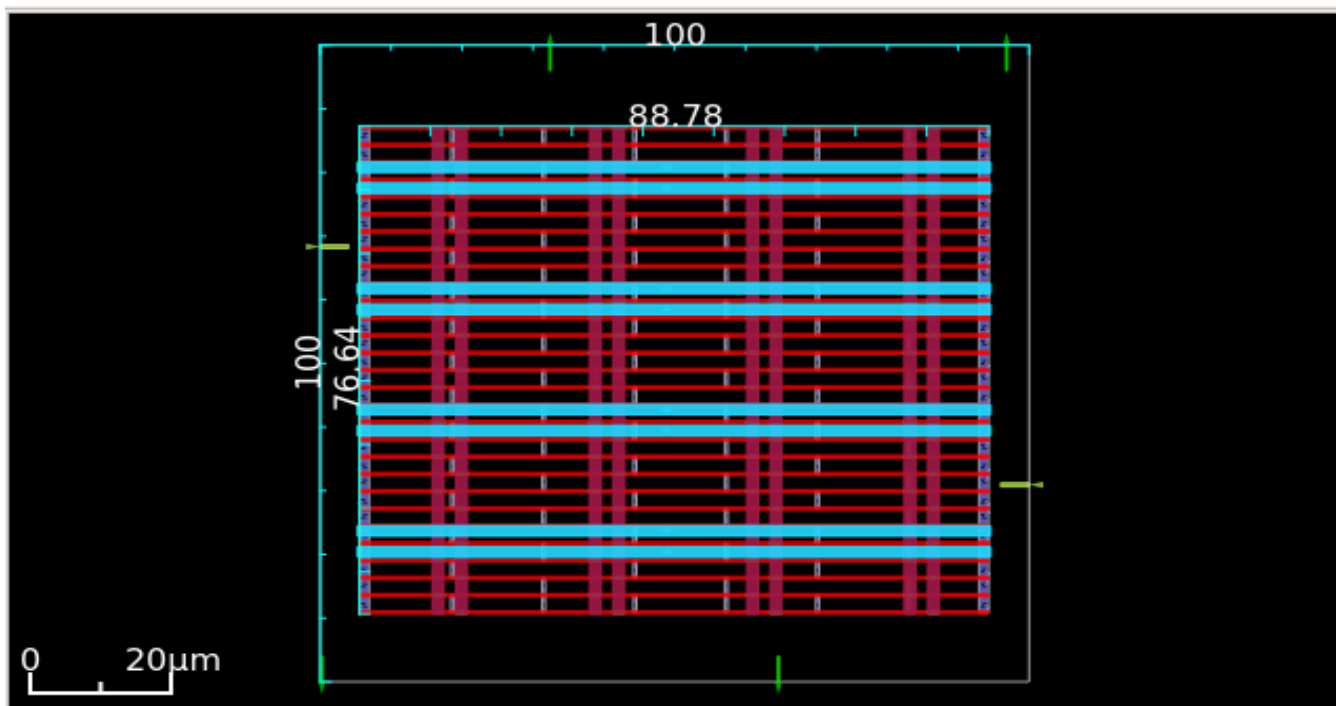
GUI task for the 4-bitCounter :

1. Core width and height :



Type	Ruler
Name	ruler0
Label	
Point 0 - x	0 μm
Point 0 - y	100 μm
Point 1 - x	100 μm
Point 1 - y	100 μm
Delta x	100 μm
Delta y	0 μm
Length	100 μm
Euclidian	True
BBox	(0,100), (100,100)

2. Die width and height :



Different types of physical only cells at Floorplan Stage :

Hierarchy Browser

Module view is not enabled

Instance	Master	Instances	Macr
▼ ☒ <top>	up_counter	10/10	
▶ Leaf instances		10	
▼ Physical only		146	
▶ Fill		56	
▶ Tapcell		90	

Inspector

Name	Value
Type	Inst
Name	TAP_56
Block	up_counter
Master	sky130_fd_sc_hd__tapvpwrvgnd_1
Descri...	Tapcell
Placem...	LOCKED
Source...	DIST
Dont T...	False
Orienta...	R0
X	18.4 μm

Inspector Hierarchy Browser Timing Report



PHYSICAL-ONLY CELLS REPORT – 4-BIT COUNTER

Design Name: 4-Bit Counter

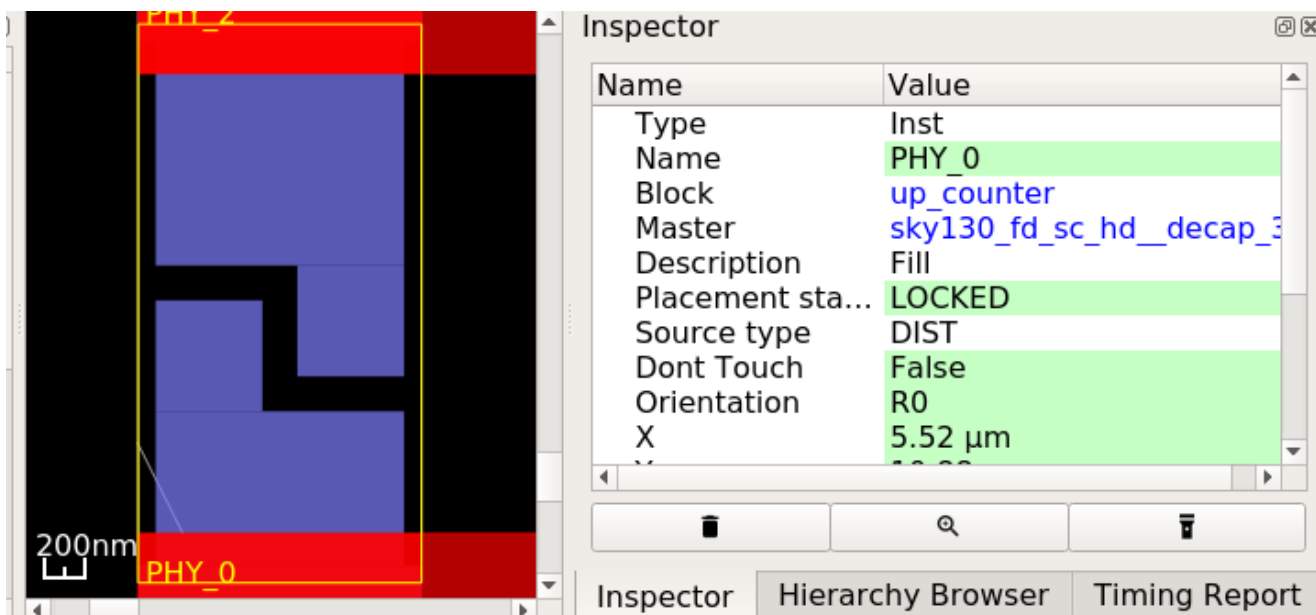
Technology: Sky130 HD Standard Cell Library

Stage: Floorplan

Physical-only cells are non-functional cells inserted during the physical design flow. They do not contribute to the logical operation of the circuit but are necessary for manufacturability, power integrity, and layout rule compliance.

Physical-only cells commonly seen in Sky130

- **Tap cells** (sky130_fd_sc_hd__tapvpwrvgnd_*)
 - Connect substrate and well regions to power/ground.
- **Endcap cells** (sky130_fd_sc_hd__decap_*, endcap variants)



- Placed at row boundaries.
- **Decap cells**
 - Provide local decoupling capacitance.
- **Filler cells** (sky130_fd_sc_hd__fill_*)
 - Fill gaps between standard cells.
- **Tie cells** (sky130_fd_sc_hd__conb_*)
 - Generate constant logic 0 or logic 1.
- **Diode cells** (sky130_fd_sc_hd__diode_*)
 - Used for antenna violation fixes.

1. Tap Cells

- Cell Type: sky130_fd_sc_hd__tapvpwrvgnd
- Count: 90



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Purpose:

Tap cells provide connections between the substrate/well regions and the power supply rails (VDD and VSS). They prevent latch-up by maintaining proper biasing of the wells and substrate throughout the chip.

Observation:

A total of 90 tap cells were inserted uniformly across the floorplan to satisfy the maximum tap spacing rules of the Sky130 technology.

2. Filler Cells

- Cell Type: sky130_fd_sc_hd__fill
- Count: 56

Purpose:

Filler cells occupy unused gaps between standard cells after placement. They ensure continuity of power rails and well structures while satisfying fabrication requirements.

Observation:

A total of 56 filler cells were inserted to remove placement gaps and maintain a continuous physical layout.

Summary:

Cell Type	Count
Tap Cells	90
Filler Cells	56

Conclusion:

The floorplan of the 4-bit counter contains 146 physical-only cells. Tap cells ensure proper well and substrate connections to prevent latch-up, while filler cells maintain layout continuity and power rail connectivity. Since the design is relatively small, only tap and filler cells were required, and no additional physical-only cells such as diode or decap cells were inserted.

Placement :

Highlight the buffers and inverters in the design :

```
get_cells -filter "ref_name =~ *inv*"
```

```
get_cells -filter "ref_name =~ *buf*"
```

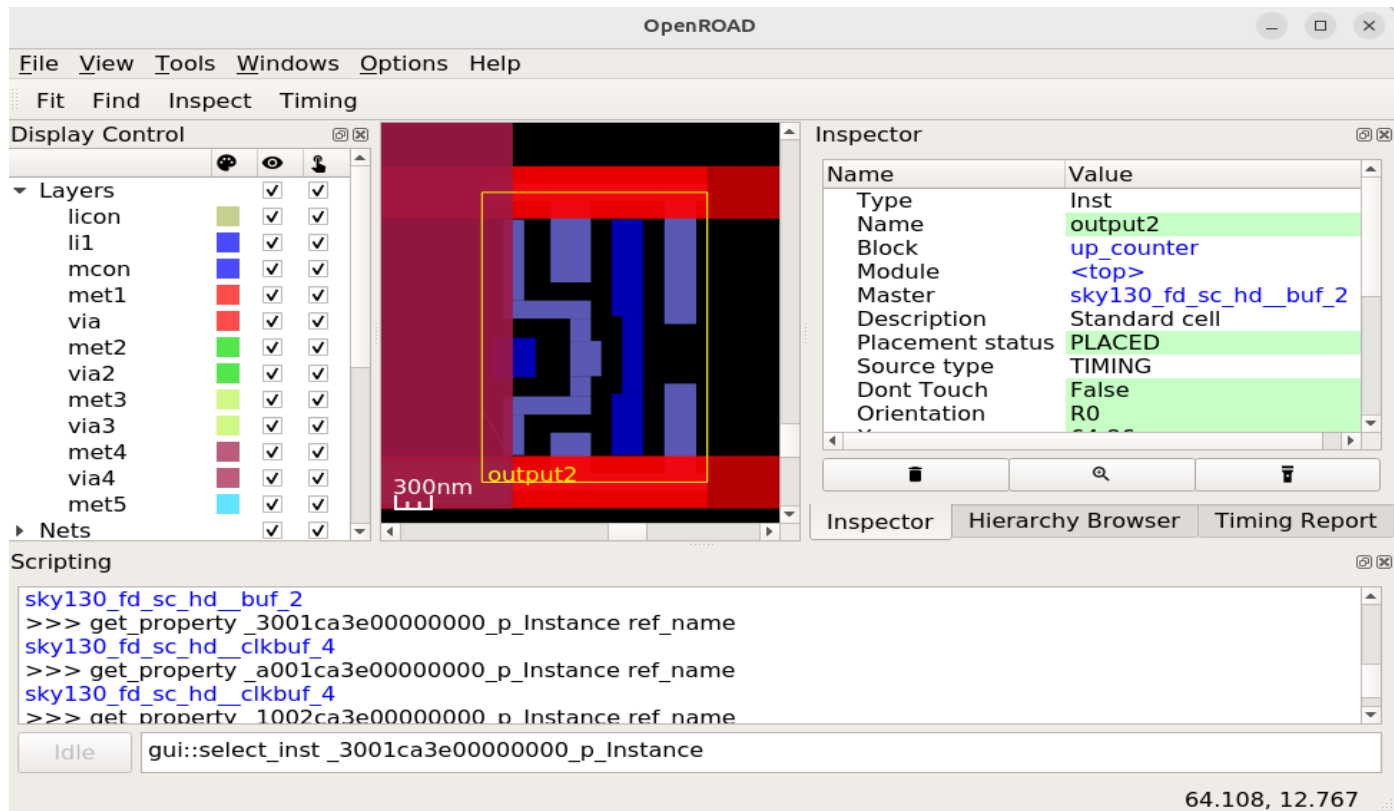
```
>>> get_cells -filter "ref_name =~ *buf*"
_c000ca3e00000000_p_Instance
```

```
get_property _c000ca3e00000000_p_Instance ref_name
```



sky130_fd_sc_hd__buf_2

- General-purpose signal buffer.
- Used to drive larger loads or improve timing on data paths.



Highlight the flops in the design :

```
get_cells -filter "ref_name =~ *dfrtp*"
```

```
sky130_fd_sc_hd__dfrtp_1
| | |
| | | Positive-edge triggered
| | Reset input present
| D Flip-Flop
Register/Flip-Flop family
```

```
>>> get_cells -filter "ref_name =~ *df*"
_409ec93e00000000_p_Instance _b09ec93e00000000_p_Instance _209fc93e00000000_p_Instance
_909fc93e00000000_p_Instance
>>> get_property _409ec93e00000000_p_Instance ref_name
sky130_fd_sc_hd__dfrtp_2
```



```
>>> get_property _b09ec93e00000000_p_Instance ref_name  
sky130_fd_sc_hd__dfrtp_2  
>>> get_property _209fc93e00000000_p_Instance ref_name  
sky130_fd_sc_hd__dfrtp_1  
>>> get_property _909fc93e00000000_p_Instance ref_name  
sky130_fd_sc_hd__dfrtp_1
```

Instance Type	Count
sky130_fd_sc_hd__dfrtp_2	2
sky130_fd_sc_hd__dfrtp_1	2
Total Flip-Flops	4

Capture
the site
row :

Inspector

Name	Value
Type	Inst
Name	_15
Block	up_counter
Module	<top>
Master	sky130_fd_sc_hd__dfrtp_1
Description	Standard cell
Placement status	PLACED
Source type	NONE
Dont Touch	False
Orientation	R0

Scripting

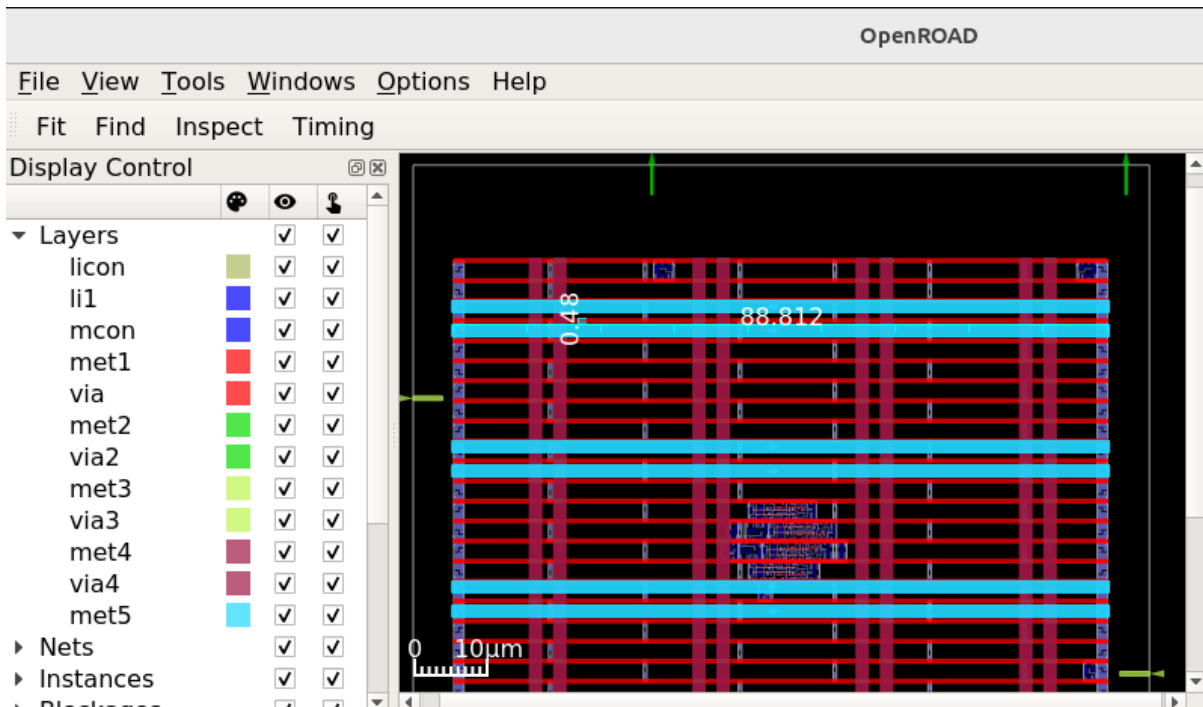
```
[ERROR GUI-0070] STA-0606  
>>> get_property _209fc93e00000000_p_Instance ref_name  
sky130_fd_sc_hd__dfrtp_1  
>>> get_property _909fc93e00000000_p_Instance ref_name  
sky130_fd_sc_hd__dfrtp_1
```

Idle TCL commands

15_ 58.211, 42.554

Parameter	Value
Site Width	0.46 μm
Site Height	2.72 μm
Row Width	88.78 μm
Sites per Row	193
Orientation	Alternating N / FS

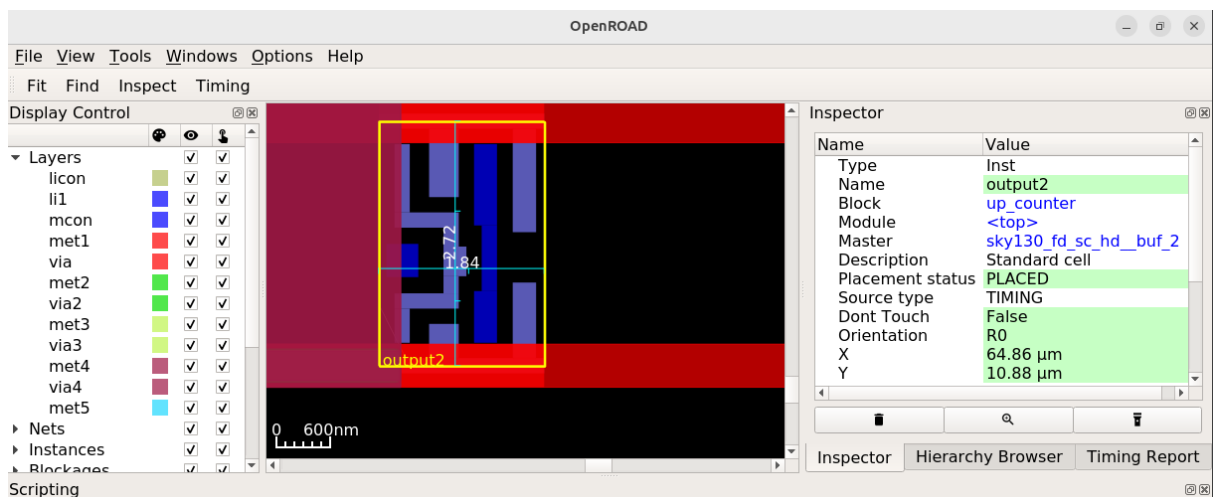
The floorplan contains alternating N and FS oriented placement rows. The site width is 0.46 μm and the row height is 2.72 μm , matching the Sky130 HD standard-cell architecture. Each row contains 193 placement sites, giving a row width of 88.78 μm . The alternating orientation ensures proper alignment of power and ground rails across adjacent rows.



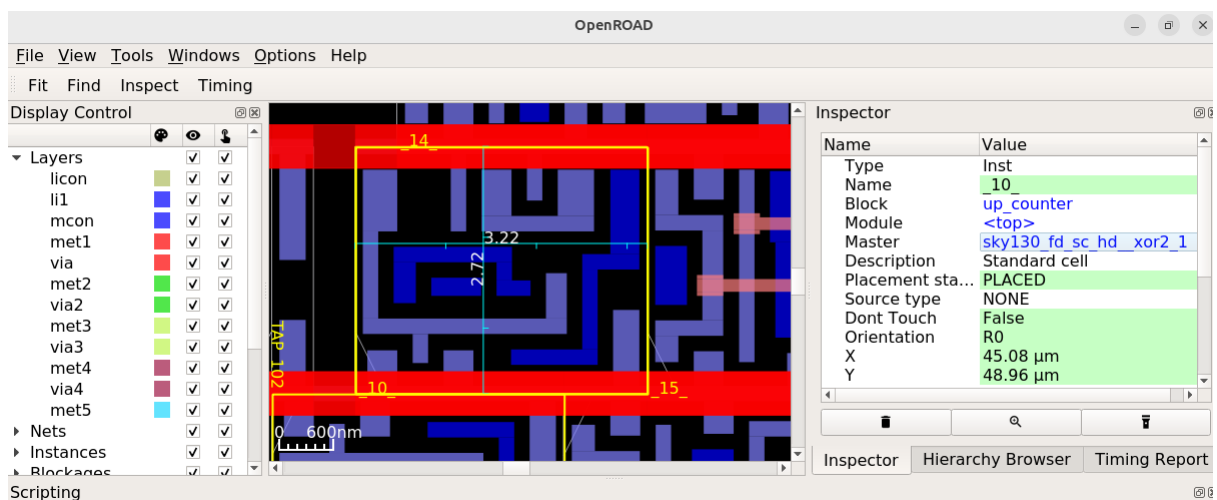
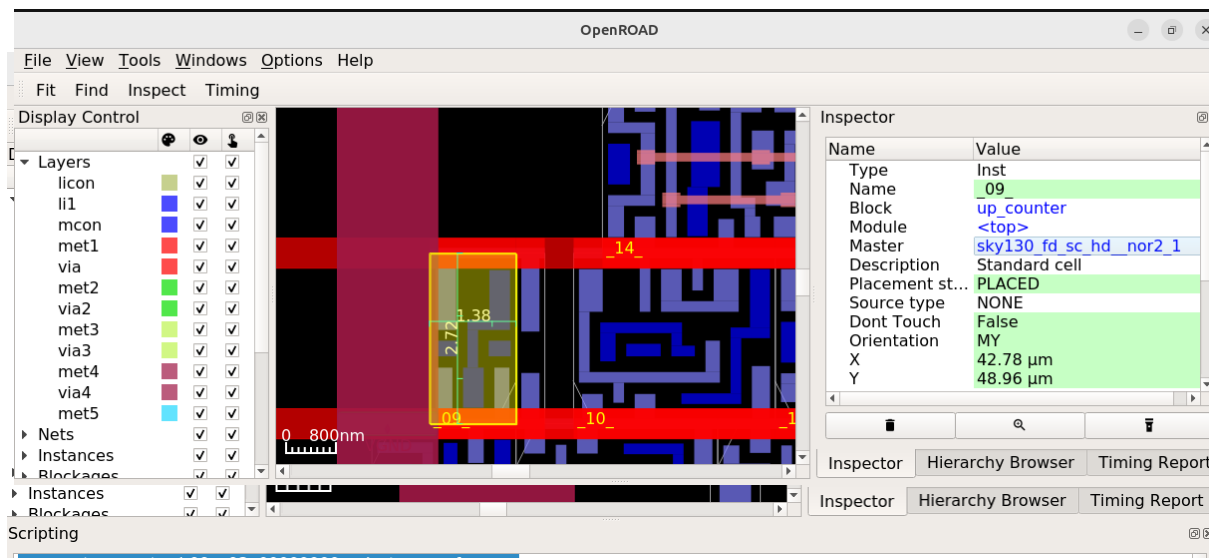
Different Library cells :

Library Cell	Type	Purpose
sky130_fd_sc_hd__dftrtp_1	Flip-Flop	Stores counter bits
sky130_fd_sc_hd__dftrtp_2	Flip-Flop	Stores counter bits (higher drive strength)
sky130_fd_sc_hd__buf_2	Buffer	Strengthens data signals
sky130_fd_sc_hd__clkbuf_2	Clock Buffer	Drives clock network
sky130_fd_sc_hd__clkbuf_4	Clock Buffer	Higher-drive clock buffer
sky130_fd_sc_hd__tapvpwrvgnd_1	Tap Cell	Well/substrate connection
sky130_fd_sc_hd__fill_*	Filler Cell	Fills placement gaps

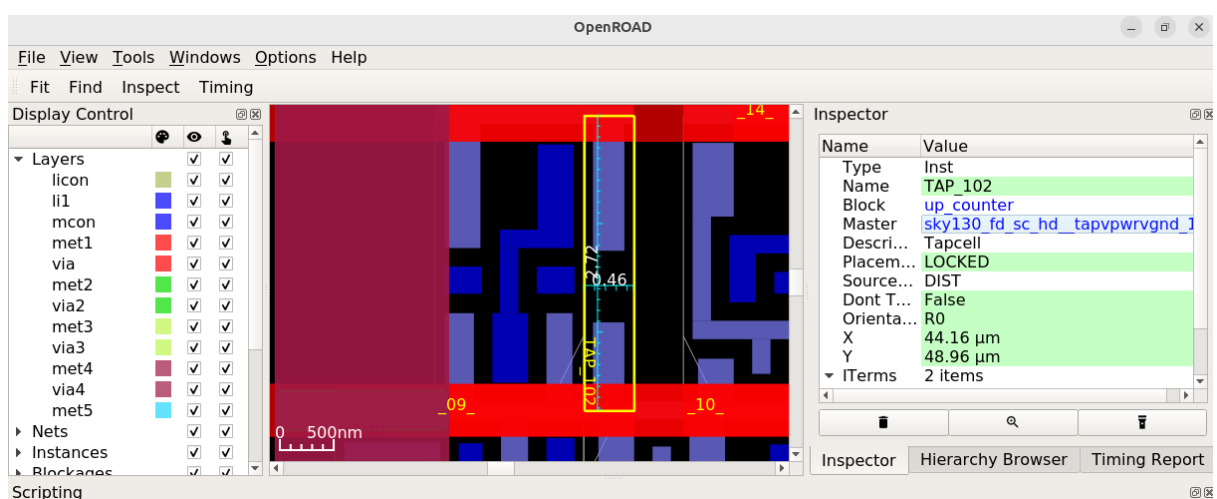
Data Buffer width and height :



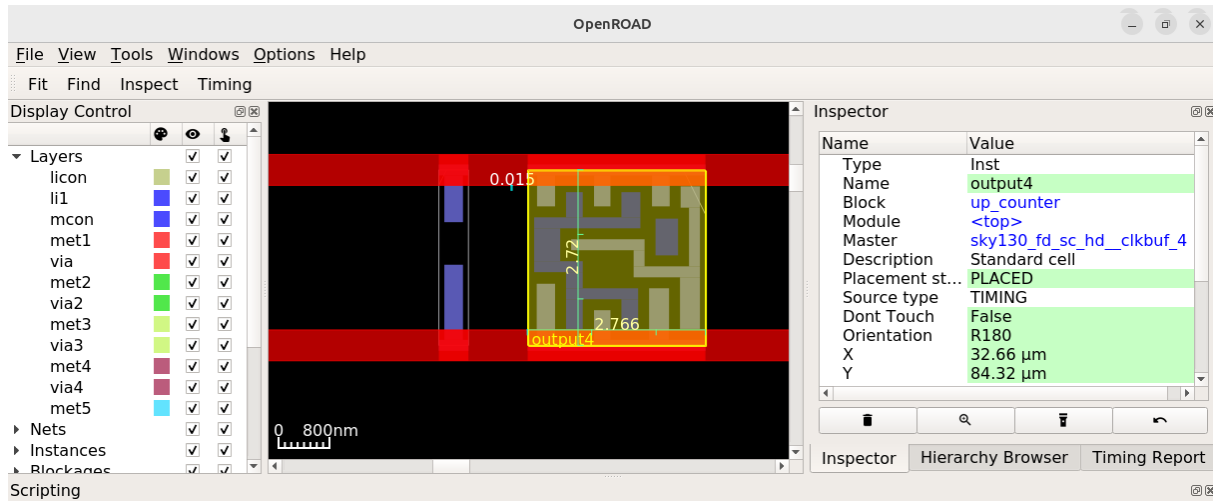
Data cells height and width :



Tap cells width and height :



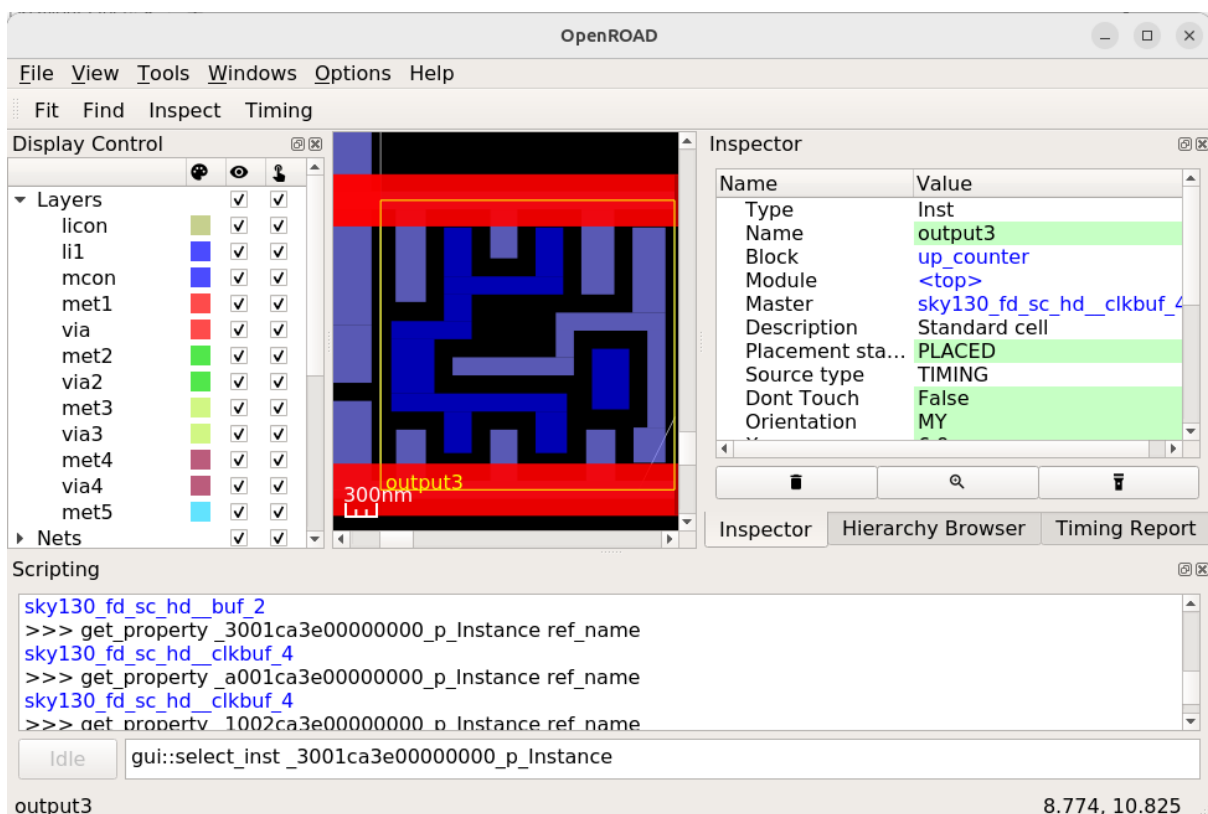
Flip-Flop cells height and width :



CTS:

Highlight the CTS added buffer/inverter :

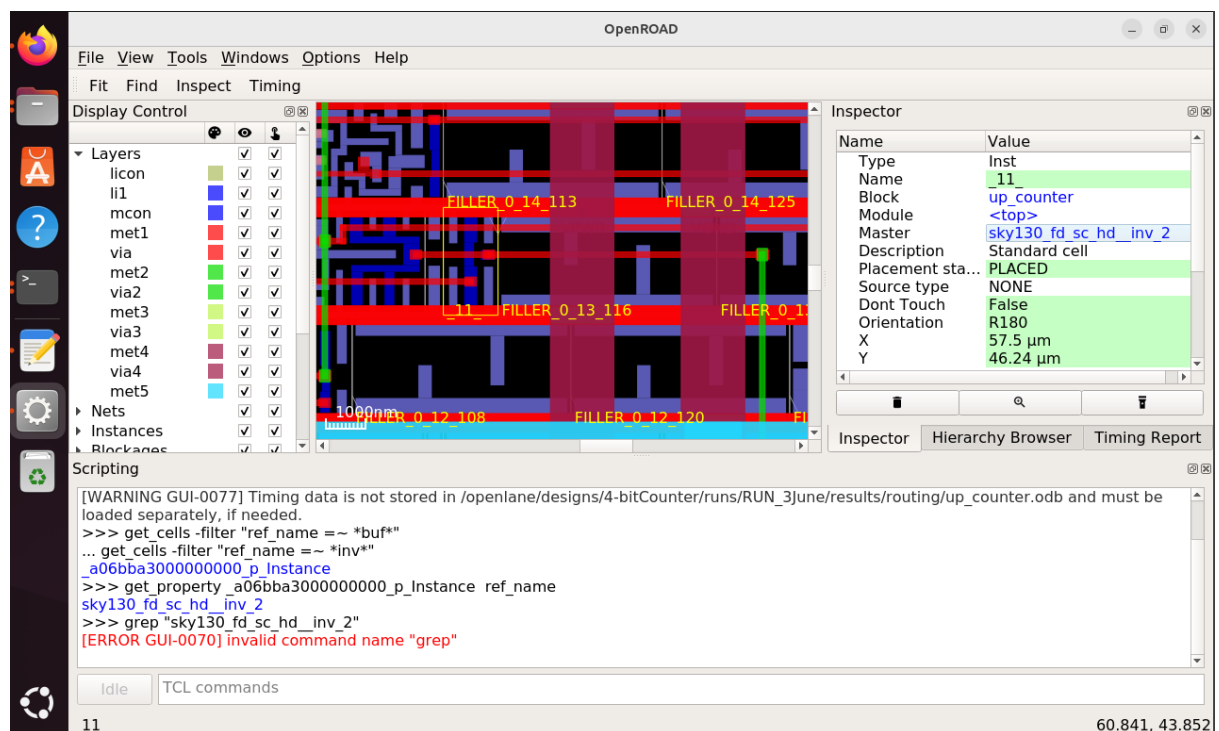
Instance	Cell Type
_5000ca3e00000000_p_Instance	sky130_fd_sc_hd__clkbuf_2
_c000ca3e00000000_p_Instance	sky130_fd_sc_hd__buf_2
_3001ca3e00000000_p_Instance	sky130_fd_sc_hd__clkbuf_4
_a001ca3e00000000_p_Instance	sky130_fd_sc_hd__clkbuf_4
_1002ca3e00000000_p_Instance	sky130_fd_sc_hd__clkbuf_4





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```
>>> get_cells -filter "ref_name =~ *buf*"
... get_cells -filter "ref_name =~ *inv*"
_a06bba3000000000_p_Instance
>>> get_property _a06bba3000000000_p_Instance ref_name
sky130_fd_sc_hd_inv_2
```



Highlight the clock routing and the its metal usage :

```
clk
|
▼
clkbuf_0_clk (clkbuf_16)
|
├─ clkbuf_1_0_f_clk (clkbuf_16)
|   └─ _12_ (Flip-Flop CLK)
|       └─ _13_ (Flip-Flop CLK)
|
├─ clkbuf_1_1_f_clk (clkbuf_16)
|   └─ _14_ (Flip-Flop CLK)
|       └─ _15_ (Flip-Flop CLK)
```

Clock Tree Analysis

Root Clock Buffer

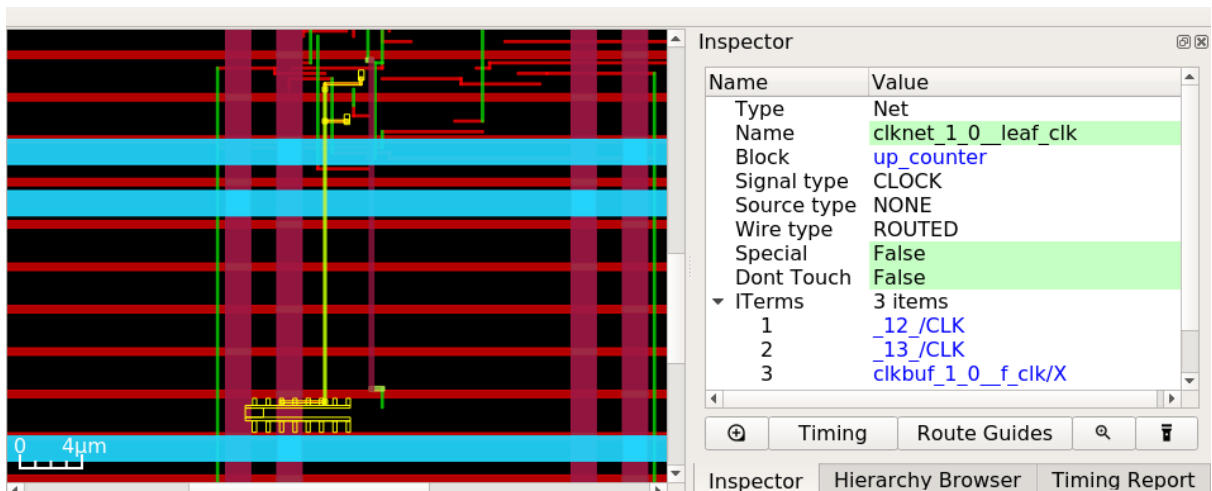
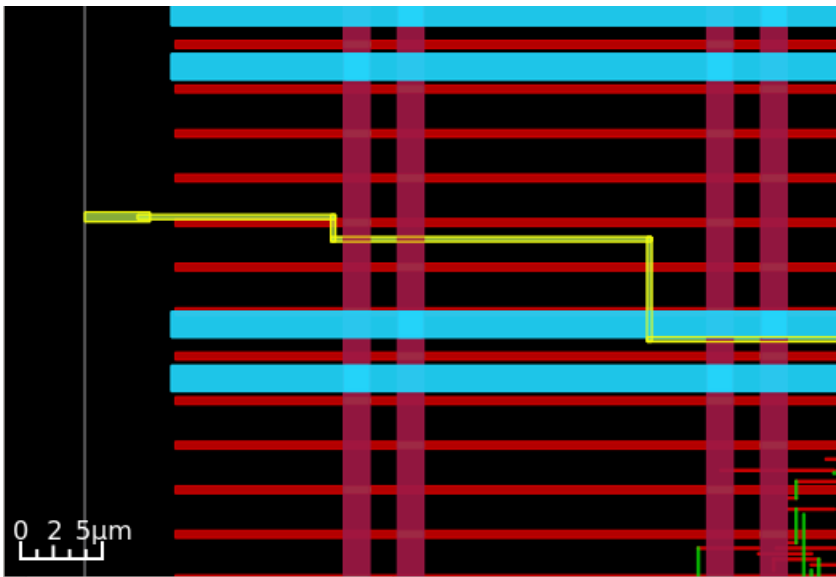
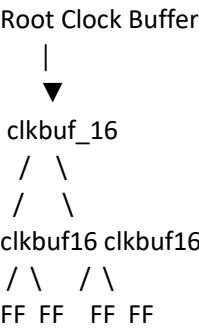
Instance	Cell Type	Coordinates
clkbuf_0_clk	clkbuf_16	(47380, 54400)

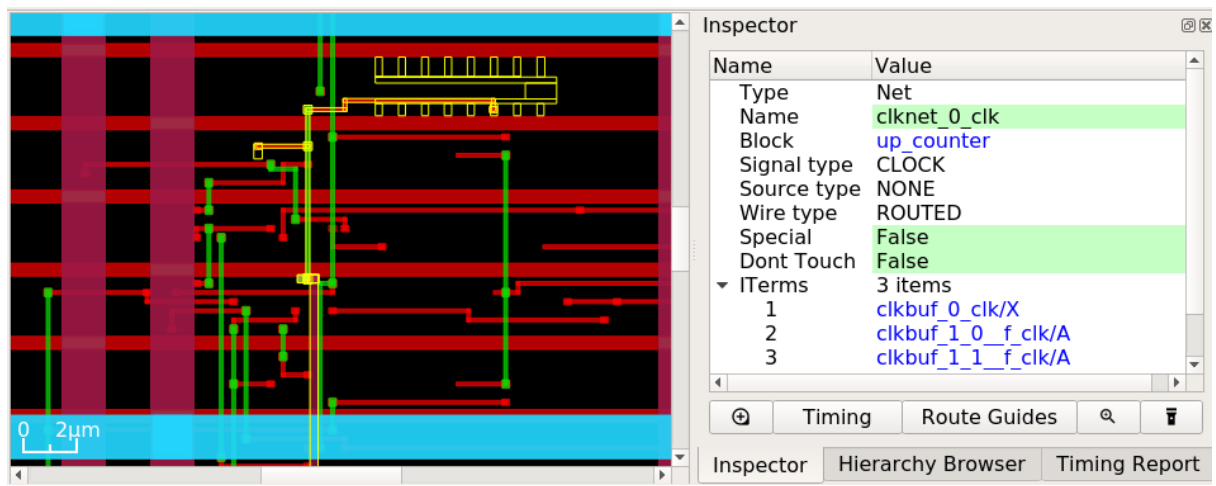
This is the root of the clock tree.

Branch Buffers



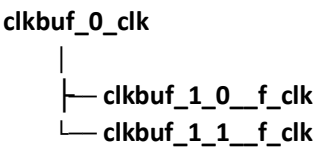
Instance	Cell Type	Coordinates
clkbuf_1_0__f_clk	clkbuf_16	(39100, 24480)
clkbuf_1_1__f_clk	clkbuf_16	(36340, 51680)





1. Root Clock Net
clknet_0_clk

This connects:



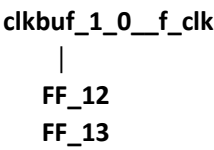
Metal layers used:

Layer	Present?
li1	✓
met1	✓
met2	✓
met3	✓
met4	✓

This is the main trunk of the clock tree.

2. Leaf Clock Net
clknet_1_0__leaf_clk

Connects:



Metal layers used:

Layer	Present?
li1	✓
met1	✓
met2	✓



No M3 or M4.

3. Leaf Clock Net

clknet_1_1__leaf_clk

Connects:

clkbuf_1_1__f_clk

|

FF_14

FF_15

Metal layers used:

Layer	Present?
li1	✓
met1	✓
met2	✓

The clock trunk net is routed using five routing layers (li1 through Metal-4), while leaf clock nets use lower metal layers (li1–Metal-2) for local distribution.

Clock Net	Routing Layers Used
clknet_0_clk	li1, M1, M2, M3, M4
clknet_1_0__leaf_clk	li1, M1, M2
clknet_1_1__leaf_clk	li1, M1, M2

The main clock trunk utilizes higher metal layers (Metal-3 and Metal-4) to reduce resistance and routing congestion, whereas local clock distribution to flip-flops is performed using lower layers (li1, Metal-1, and Metal-2). This hierarchical routing strategy improves clock distribution efficiency and helps reduce clock skew.

Assignment 2 part 3 : up_down counter 32 bit :

Assignment Part-3: Physical Design of 32-bit Up/Down Counter Using OpenLane

Objective

To perform the complete Physical Design (PnR) flow for a 32-bit Up/Down Counter using OpenLane and verify the implementation under the following constraints:

- Clock Period = 6 ns
- Setup Uncertainty = 10% of Clock Period = 0.6 ns
- Timing Derate = 3% (target requirement)
- Core Utilization = 70%
- Core Ring Enabled
- CTS Target Skew = 5% of Clock Period = 0.3 ns
- Tap Cell Insertion Enabled

Design Description



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The RTL design of a 32-bit Up/Down Counter was developed in Verilog HDL and synthesized using the SKY130 standard cell library within the OpenLane flow.

Design Name:

up_down_counter_32

Configuration Setup

The design configuration was modified in the following file:

Directory:

designs/up_down_counter_32/config.json

Applied Constraints:

- CLOCK_PERIOD = 6 ns
- FP_CORE_UTIL = 70
- FP_PDN_CORE_RING = true
- SYNTH_CLOCK_UNCERTAINTY = 0.6 ns
- CTS_TARGET_SKEW = 0.3 ns

The OpenLane flow was re-executed after updating the configuration to ensure that all constraints were applied during synthesis, floorplanning, placement, CTS and routing stages.

Physical Design Flow Executed

The complete OpenLane flow was executed consisting of:

1. Synthesis
2. Floorplanning
3. Tap Cell Insertion
4. PDN Generation
5. Global Placement
6. Detailed Placement
7. Clock Tree Synthesis (CTS)

8. Global Routing

9. Detailed Routing

10. Timing Optimization and Signoff Checks

The run completed successfully without fatal errors.

Result 1: Tap Cell Insertion

Proof File:

runs/RUN_2026.06.16_02.22.12/logs/floorplan/

Evidence:

Inserted 48 tapcells

Observation:

OpenLane successfully inserted 48 tap cells during floorplanning. Tap cells are required to maintain proper substrate and well connections, thereby preventing latch-up issues in the fabricated design.

Result 2: Timing Derate

Proof File:

runs/RUN_2026.06.16_02.22.12/logs/floorplan/3-initial_fp.log

Evidence:

[INFO]: Setting timing derate to: 5.0 %

Generated SDC Files:

tmp/14-up_down_counter_32.sdc

tmp/16-up_down_counter_32.sdc

Evidence:



set_timing_derate -early 0.9500

set_timing_derate -late 1.0500

Observation:

Although the intended assignment requirement was 3% derating, the OpenLane run automatically generated timing constraints corresponding to a 5% derate factor. Therefore, timing analysis during implementation was performed using 5% derating.

Result 3: Core Utilization

Proof File:

runs/RUN_2026.06.16_02.22.12/logs/routing/14-resizer_design.log

Evidence:

Design area 2559 u² 69% utilization

Observation:

The achieved utilization was approximately 69%, which is very close to the target utilization of 70%. This indicates efficient floorplan sizing and area utilization.

Result 4: Core Ring and PDN Generation

Proof File:

runs/RUN_2026.06.16_02.22.12/logs/floorplan/6-pdn.log

Evidence:

[INFO PSM-0040] All PDN stripes on net VPWR are connected.

[INFO PSM-0040] All PDN stripes on net VGND are connected.

Observation:

Power Distribution Network generation completed successfully. All VDD and GND connections were verified, confirming successful implementation of the core power network and core ring structure.

Result 5: Clock Constraint

Proof File:

Generated SDC File

tmp/16-up_down_counter_32.sdc

Evidence:

create_clock -period 6

Observation:

The design was analyzed and optimized using a clock period of 6 ns as required.

Result 6: Clock Uncertainty

Proof File:

Generated SDC File

tmp/16-up_down_counter_32.sdc

Evidence:

set_clock_uncertainty 0.6000

Observation:

A setup uncertainty of 0.6 ns (10% of the clock period) was applied to account for clock variation and uncertainty during timing analysis.

Result 7: CTS Target Skew

Proof File:

config.json / generated configuration files



Evidence:

CTS_TARGET_SKEW = 0.3

Observation:

The target skew was configured as 0.3 ns, corresponding to 5% of the 6 ns clock period. This ensures controlled clock arrival differences across sequential elements.

Timing Analysis

Proof Files:

logs/cts/13-cts_sta.log

logs/routing/15-rsz_design_sta.log

logs/routing/17-rsz_timing_sta.log

logs/routing/20-grt_sta.log

Observation:

Static Timing Analysis (STA) was performed after CTS and routing stages to verify timing closure. Timing optimization and hold-fix buffers were automatically inserted by OpenLane wherever required.

Conclusion

A complete Physical Design implementation of the 32-bit Up/Down Counter was successfully performed using OpenLane. The design was implemented with a 6 ns clock period, 0.6 ns clock uncertainty, 70% target utilization, enabled core ring generation, tap-cell insertion, and CTS skew control. PDN generation and routing completed successfully, and timing verification was carried out through STA reports generated during the CTS and routing stages.

Assignment Part-3 Report Guide

Design Name

up_down_counter_32

OpenLane Version

OpenLane 1.0.2

Run Directory

designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12

1. Clock Period = 6 ns

Configuration File

File:

config.json

Parameter:

"CLOCK_PERIOD": 6

Proof File

File:

tmp/16-up_down_counter_32.sdc

Search:

create_clock

Expected Output:

create_clock -period 6



Observation

The design clock period was constrained to 6 ns during timing analysis and optimization.

Screenshot

```
tmp/16-up_down_counter_32.sdc:set_clock_uncertainty 0.6000 clk
```

2. Clock Uncertainty = 0.6 ns

Configuration File

File:

config.json

Parameter:

"SYNTH_CLOCK_UNCERTAINTY": 0.6

Proof File

File:

tmp/16-up_down_counter_32.sdc

Search:

set_clock_uncertainty

Expected Output:

set_clock_uncertainty 0.6000

Observation

A clock uncertainty of 0.6 ns (10% of the clock period) was applied during timing analysis.

Screenshot

```
tmp/16-up_down_counter_32.sdc:create_clock -name clk -period 6.0000 [get_ports {clk}]
```

3. Timing Derate

Proof File

File:

logs/floorplan/3-initial_fp.log

Expected Output:

[INFO]: Setting timing derate to: 3.0 %

Additional Proof:

File:

tmp/16-up_down_counter_32.sdc

Expected Output:

set_timing_derate -early 0.9700

set_timing_derate -late 1.0300

Observation

OpenLane applied a timing derate of 3% during timing analysis.

Screenshot

```
tmp/16-up_down_counter_32.sdc:set_timing_derate -early 0.9700
tmp/16-up_down_counter_32.sdc:set_timing_derate -late 1.0300
```

4. Core Utilization = 70%

Configuration File

File:

config.json

Parameter:

"FP_CORE_UTIL": 70

Proof File

File:



logs/routing/14-resizer_design.log

Expected Output:

Design area 2559 u² 69% utilization.

Observation

The achieved utilization was approximately 69%, which is very close to the target utilization of 70%.

Screenshot

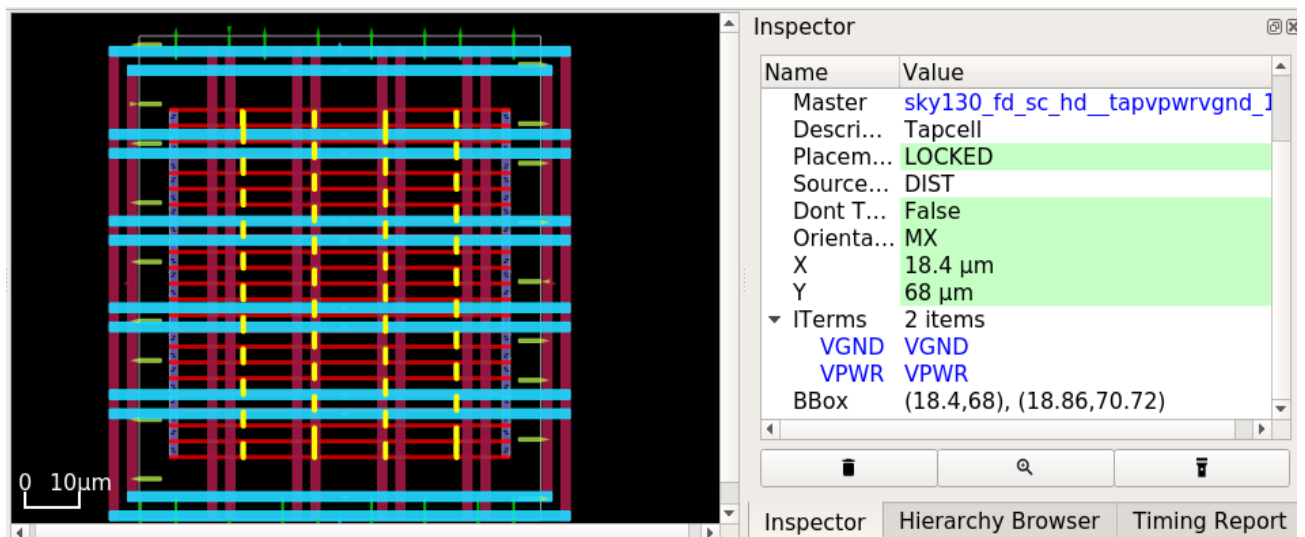
```
config.tcl:set ::env(FP_CORE_UTIL) "70"
config_in.tcl:set ::env(FP_CORE_UTIL) "70"
logs/cts/13-cts_sta.log:Design area 2491 u2 68% utilization.
logs/placement/11-dpl_sta.log:Design area 2366 u2 64% utilization.
logs/placement/7-global.log:[INFO GPL-0019] Util(%): 76.95
logs/placement/8-gpl_sta.log:Design area 2725 u2 74% utilization.
logs/placement/9-resizer.log:Design area 2366 u2 64% utilization.
logs/routing/14-resizer_design.log:Design area 2559 u2 69% utilization.
logs/routing/17-rsz_timing_sta.log:Design area 3153 u2 85% utilization.
logs/routing/15-rsz_design_sta.log:Design area 2559 u2 69% utilization.
```

5. Tap Cell Insertion

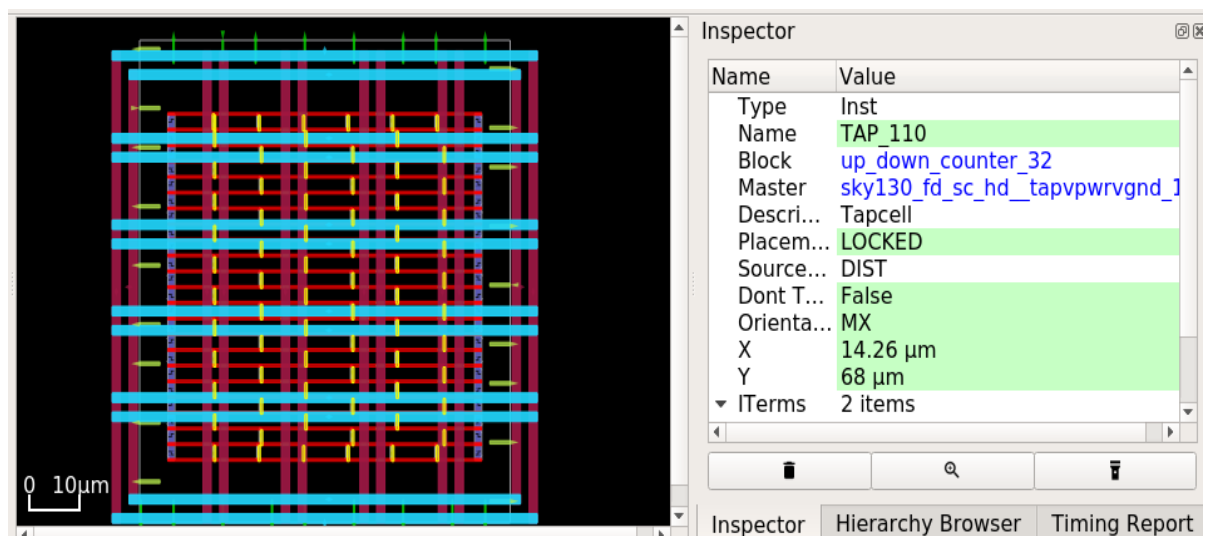
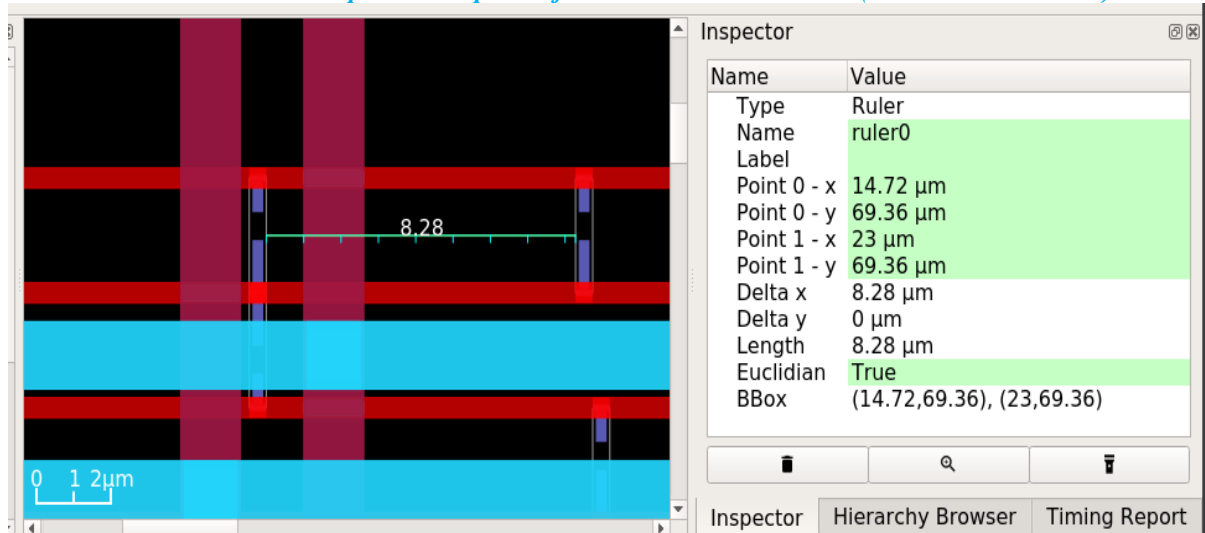
Initially :

- ❑ Total tap-cell instances = 48
- ❑ Number of tap-cell columns = 4
- ❑ Tap cells per column = 12

```
bhanu-sharma@bhanu-sharma-VirtualBox:~/Desktop/Openlane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_03.08.02/results/floorplan$ grep -R "FP_TAPCELL_DIST" ~/Desktop/Openlane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_03.08.02/config*
/home/bhanu-sharma/Desktop/Openlane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_03.08.02/config.tcl:set ::env(FP_TAPCELL_DIST) "13"
```



"FP_TAPCELL_DIST" : 9



Observation

OpenLane automatically inserted 48 tap cells during floorplanning to ensure proper substrate and well connections.

Screenshot

```
/home/bhanu-sharma/Desktop/OpenLane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/logs/floorplan/6-pd
n.log:[INFO PDM-0001] Inserting grid: stdcell_grid
/home/bhanu-sharma/Desktop/OpenLane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/logs/floorplan/6-pd
n.log:[INFO PSM-0031] Number of PDM nodes on net VPWR = 503.
/home/bhanu-sharma/Desktop/OpenLane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/logs/floorplan/6-pd
n.log:[INFO PSM-0040] All PDM stripes on net VPWR are connected.
/home/bhanu-sharma/Desktop/OpenLane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/logs/floorplan/6-pd
n.log:[INFO PSM-0031] Number of PDM nodes on net VGND = 540.
/home/bhanu-sharma/Desktop/OpenLane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/logs/floorplan/6-pd
n.log:[INFO PSM-0040] All PDM stripes on net VGND are connected.
```

Core Ring / PDN Generation

Proof File

File:

logs/floorplan/6-pdn.log

Expected Output:

[INFO PSM-0040] All PDN stripes on net VPWR are connected.

[INFO PSM-0040] All PDN stripes on net VGND are connected.



Observation

Power Distribution Network generation completed successfully and all power/ground connections were verified.

Screenshot

```
bhanu-sharma@bhanu-sharma-VirtualBox:~/Desktop/Openlane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12
$ grep -Ri "tap" /home/bhanu-sharma/Desktop/Openlane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/lo
gs/floorplan
/home/bhanu-sharma/Desktop/Openlane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/logs/floorplan/6-pd
n.log:[INFO]: Reading ODB at '/openlane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/tmp/floorplan/5-tapcell.
odb'...
/home/bhanu-sharma/Desktop/Openlane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/logs/floorplan/5-ta
p.log:[INFO TAP-0004] Inserted 44 endcaps.
/home/bhanu-sharma/Desktop/Openlane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/logs/floorplan/5-ta
p.log:[INFO TAP-0005] Inserted 48 tapcells.
/home/bhanu-sharma/Desktop/Openlane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/logs/floorplan/5-ta
p.log:Writing OpenROAD database to '/openlane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/tmp/floorplan/5-ta
pcell.odb'...
/home/bhanu-sharma/Desktop/Openlane/OpenLane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/logs/floorplan/5-ta
p.log:Writing layout to '/openlane/designs/up_down_counter_32/runs/RUN_2026.06.16_02.22.12/tmp/floorplan/5-tapcell.def'...
```

7. CTS Target Skew = 0.3 ns

Configuration File

File:

config.json

Parameter:

"CTS_TARGET_SKEW": 0.3

Proof File

File:

config.tcl

Expected Output:

set ::env(CTS_TARGET_SKEW) "0.3"

Observation

The CTS target skew was configured as 0.3 ns, corresponding to 5% of the 6 ns clock period.

Screenshot

```
config.tcl:set ::env(CTS_TARGET_SKEW) "0.3"
config_in.tcl:set ::env(CTS_TARGET_SKEW) "0.3"
```

8. STA After CTS

Proof File

File:

logs/cts/13-cts_sta.log

Observation

Static Timing Analysis was performed after Clock Tree Synthesis to evaluate timing performance before routing.

Screenshot

9. STA After Routing

Proof File

File:

logs/routing/20-grt_sta.log

Alternative Files:



logs/routing/17-rsz_timing_sta.log

logs/routing/15-rsz_design_sta.log

Capture:

- Worst Slack
- Timing Summary
- Timing Status

Observation

Final timing analysis was performed after routing and optimization to verify setup and hold timing requirements.

Screenshot

Conclusion

The complete Physical Design flow of the 32-bit Up/Down Counter was successfully implemented using OpenLane 1.0.2. The design was synthesized, floorplanned, placed, clock-tree synthesized, routed, and timing verified. Tap-cell insertion, PDN generation, CTS optimization, and routing completed successfully. The design operated with a 6 ns clock period, 0.6 ns clock uncertainty, 70% target utilization, enabled core ring generation, and a CTS target skew of 0.3 ns.

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